Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [20EE32]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ELECTRONIC DEVICES AND CIRCUITS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Explain the purpose of a DC load line using a plot. Write the equation for a drawing a dc load line for a series circuit consisting of a supply voltage (E), a resistor (R_1) and a diode (D_1). **07**

b. Construct a double diode clipper which limits at two independent levels and explain its working. **07**

2. a. Explain with the help of a circuit diagram the working of a full wave bridge rectifier. Derive expressions for **07**

i. V_{dc}
ii. Rectification efficiency
iii. Ripple factor

b. With a neat block diagram explain the working series voltage regulator and list the advantages **07**

3. a. Show a block diagram of a biased npn transistor. Describe the depletion region, barrier voltages, and the movement of charge carriers across the junctions. Explain the device operation. **06**

b. Illustrate and explain the input and output characteristics of a npn transistor for Common-base configuration. **08**

OR

4. a. Discuss the switching characteristics of a BJT with the help of its equivalent circuit and waveforms. **07**

b. Construct a collector-to-base bias circuit using an NPN transistor. Show all voltage polarities and current directions with proper analysis. **07**

5. a. With necessary sketches, explain the switching characteristics of an MOSFET. **08**

b. List the difference between MOSFET and JFET **06**

OR

6.	**a.**	Explain with the help of neat diagram construction, working & VI characteristics of n channel JFET.	**08**
	b.	Mention the applications of FET.	**04**
	c.	Define drain resistance?	**02**
7.	**a.**	With neat circuit diagram explain the working of phase shift oscillator	**08**
	b.	With neat circuit diagram explain the working of Wien bridge oscillator	**06**

OR

8.	**a.**	With neat circuit diagram explain the working of class A amplifier	**07**
	b.	With neat circuit diagram explain the working of Class AB amplifier	**07**